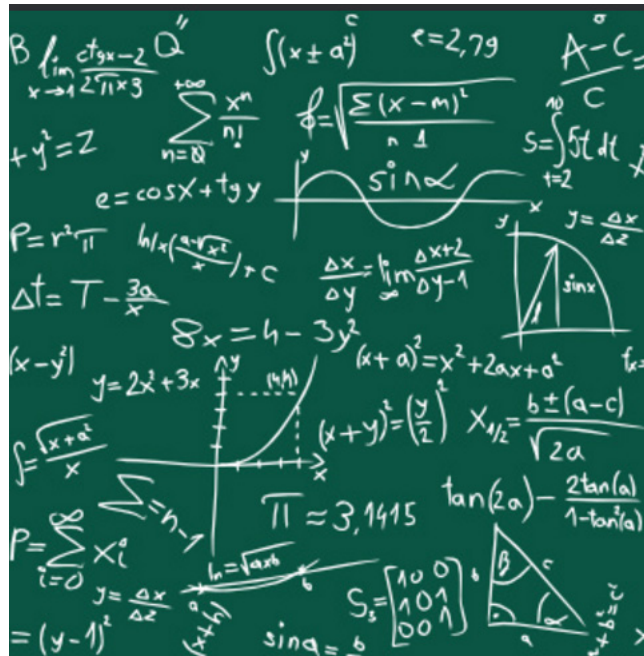


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Exercises: equations and inequalities in one real variable



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “*Esercitazioni Analisi matematica 1*”, Marco Bramanti, Esculapio.

Exercise 1

Solve the following inequality in the real variable x :

$$\frac{x^2 - 2x}{x^2 - 4x + 3} > 0$$

The numerator $x^2 - 2x$ is positive for $x < 0$ or for $x > 2$, zero for $x = 0$, $x = 2$, and negative for $0 < x < 2$. The denominator $x^2 - 4x + 3$ is positive for $x < 1$ or for $x > 3$, and negative for $1 < x < 3$. The quotient is defined and has the required positive sign for

$$x \in (-\infty, 0) \cup (1, 2) \cup (3, +\infty).$$

Exercise 2

Solve the following inequality in the real variable x :

$$\log(x-1)^2 - \log(x-2) > 0$$

The function $f(x) = \log(x-1)^2 - \log(x-2)$ is defined for $x > 2$, since for such x , and only for such x , we have $x-2 > 0$, $(x-1)^2 > 0$. In its domain, we have, as required, $f(x) > 0$ if and only if

$$\log(x-1)^2 > \log(x-2), \quad x > 2$$

hence if and only if

$$\begin{cases} (x-1)^2 > x-2 \\ x > 2 \end{cases}$$

We obtain

$$\begin{cases} x^2 - 3x + 3 > 0 \\ x > 2 \end{cases}$$

which is equivalent to

$$x > 2$$

since $x^2 - 3x + 3 > 0$ is satisfied for every x . On its whole domain $(2, +\infty)$ the function $f(x)$ takes positive values.

Exercise 3

Solve the following inequality in the real variable x :

$$e^x + e^{-x} < \frac{10}{3}$$

For every $x \in \mathbb{R}$

$$e^x + e^{-x} < \frac{10}{3}$$

is equivalent to

$$3e^{2x} - 10e^x + 3 < 0$$

hence to

$$\frac{1}{3} < e^x < 3.$$

It follows that the inequality is solved by

$$-\log 3 < x < \log 3.$$

Exercise 4

Solve the following inequality in the real variable x :

$$8^{x+1} \geq 2^{x^2}$$

For every $x \in \mathbb{R}$

$$8^{x+1} \geq 2^{x^2}$$

is equivalent to

$$2^{3x+3} \geq 2^{x^2}$$

hence to

$$3x + 3 \geq x^2$$

since 2^x is a strictly increasing function on $\mathbb{R}$. It follows that the inequality is solved by

$$\frac{3 - \sqrt{21}}{2} \leq x \leq \frac{3 + \sqrt{21}}{2}.$$

Exercise 5

Solve the following inequality in the real variable x :

$$\log(1 - \sin x) \geq 0$$

The function $f(x) = \log(1 - \sin x)$ is defined for $\sin x < 1$, hence for

$$x \neq \frac{\pi}{2} + 2k\pi, \quad k \in \mathbf{Z}.$$

For such x we have $f(x) \geq 0$ if and only if

$$1 - \sin x \geq 1$$

hence for

$$\sin x \leq 0.$$

The given inequality is solved for

$$\pi + 2k\pi \leq x \leq 2\pi + 2k\pi, \quad k \in \mathbf{Z}$$

i.e., for

$$x \in \bigcup_{k \in \mathbf{Z}} [\pi + 2k\pi, 2\pi + 2k\pi].$$

Exercise 6

Solve the following inequality in the real variable x :

$$|x|\sqrt{1 - 2x^2} > 2x^2 - 1$$

$\sqrt{1 - 2x^2}$ is defined for $1 - 2x^2 \geq 0$. When $1 - 2x^2 = 0$ the inequality is not satisfied, since it reduces to $0 > 0$. When $1 - 2x^2 > 0$ the inequality is satisfied, since the left-hand side $|x|\sqrt{1 - 2x^2}$ is the product of two positive numbers while the right-hand side $2x^2 - 1 = -(1 - 2x^2)$ is negative. Therefore the inequality is solved for

$$-\sqrt{\frac{1}{2}} < x < \sqrt{\frac{1}{2}}.$$

Exercise 7

Solve the following equation in the real variable x :

$$\sin^2 x = 2 \cos^2 x - \frac{1}{2}$$

The equation

$$\sin^2 x = 2 \cos^2 x - \frac{1}{2}$$

is equivalent to

$$2 \sin^2 x = 4(1 - \sin^2 x) - 1$$

$$6 \sin^2 x = 3$$

$$\sin x = \pm \sqrt{\frac{1}{2}}.$$

The solutions are given by

$$x = \frac{\pi}{4} + \frac{k\pi}{2}, \quad k \in \mathbf{Z}.$$

Exercise 8

Solve the following inequality in the real variable x :

$$e^{\sin^2 x - \sin x} \leq 1$$

For every $x \in \mathbb{R}$, $e^{\sin^2 x - \sin x} \leq 1$ is equivalent to

$$\sin^2 x - \sin x \leq 0$$

hence to

$$0 \leq \sin x \leq 1.$$

The solutions are given by

$$2k\pi \leq x \leq \pi + 2k\pi, \quad k \in \mathbf{Z}$$

i.e., by

$$x \in \bigcup_{k \in \mathbf{Z}} [2k\pi, \pi + 2k\pi].$$

Exercise 9

Solve the following inequality in the real variable x :

$$2|x^2 - x| > |x|$$

The inequality $2|x^2 - x| > |x|$ is not satisfied for $x = 0$. For $x \neq 0$, dividing by the positive term $|x|$, it is equivalent to

$$2|x - 1| > 1$$

hence to

$$|x - 1| > 1/2.$$

The solutions are given by

$$(x - 1 < -1/2, x \neq 0) \vee (x - 1 > 1/2),$$

$$(x < 1/2, x \neq 0) \vee x > 3/2$$

hence by

$$x \in (-\infty, 0) \cup (0, 1/2) \cup (3/2, +\infty).$$

Exercise 10

Solve the following inequality in the real variable x :

$$(x + 1)^{x^2 - 1} > 1$$

The function $f(x) = (x + 1)^{x^2 - 1}$ is defined for $x > -1$. For such x , $f(x) > 1$ is equivalent to

$$e^{(x^2 - 1)\log(x + 1)} > 1$$

hence to

$$(x^2 - 1)\log(x + 1) > 0, \quad x > -1.$$

Dividing by the positive term $x + 1$ we obtain

$$(x - 1)\log(x + 1) > 0, \quad x > -1.$$

The factor $x - 1$ is positive for $x > 1$, zero for $x = 1$, and negative for $-1 < x < 1$. The factor $\log(x + 1)$ is positive for $x > 0$, zero for $x = 0$, and negative for $-1 < x < 0$. By the rule of signs, the solutions are given by

$$x \in (-1, 0) \cup (1, +\infty).$$

Exercise 11

Solve the following equation in the real variable x :

$$3^{|x^2-4|} = 0$$

$3^{|x^2-4|} = 0$ has no solutions, since $3^y > 0$ for every $y \in \mathbb{R}$.

Exercise 12

Solve the following inequality in the real variable x :

$$\frac{\log_a(4x-3)}{\log_a(2x-1)} > 1 \quad 0 < a < 1$$

Because of the existence conditions of the logarithm and of the denominator, we must impose the system

$$\begin{cases} 4x-3 > 0 \\ 2x-1 > 0 \\ \log_a(2x-1) \neq 0 \end{cases}$$

which gives $\frac{3}{4} < x < 1, x > 1$. Now moving everything to the left-hand side, finding the least common denominator and using the properties of logarithms, we obtain

$$\frac{\log_a\left(\frac{4x-3}{2x-1}\right)}{\log_a(2x-1)} > 0$$

Studying the sign of the numerator and of the denominator, and keeping in mind that $a < 1$, for the numerator we obtain

$$\log_a\left(\frac{4x-3}{2x-1}\right) > 0 \iff \frac{4x-3}{2x-1} < 1 \iff \frac{1}{2} < x < 1$$

while for the denominator we obtain

$$\log_a(2x-1) > 0 \iff x < 1$$

Combining the signs, the inequality is satisfied for $\frac{1}{2} < x < 1, x > 1$. Intersecting this set of solutions with the existence conditions studied above, we obtain that the inequality is satisfied for every x belonging to the set $(\frac{3}{4}; 1) \cup (1, +\infty)$.

Exercise 13

Solve the following inequality in the real variable x :

$$\sqrt{x^2 - 3x + 5} \leq x + 3$$

This irrational inequality has the form $\sqrt{A(x)} \leq B(x)$, with $A(x)$ and $B(x)$ polynomials in x . First of all, we must impose the existence condition of the root, i.e., $x^2 - 3x + 5 \geq 0$. Moreover, since a square root is always positive or zero in $\mathbb{R}$, we impose $x + 3 \geq 0$. We therefore solve the system:

$$\begin{cases} x^2 - 3x + 5 \geq 0 \\ x + 3 \geq 0 \\ x^2 - 3x + 5 \leq (x + 3)^2 \end{cases}$$

where, in the last inequality, we were allowed to square both sides, since they are positive. The system reduces to:

$$\begin{cases} \forall x \in \mathbb{R} \\ x \geq -3 \\ x \geq -\frac{4}{9} \end{cases}$$

whose set of solutions is the interval $[-\frac{4}{9}, +\infty)$.

Exercise 14

Solve the following rational inequality in the real variable x :

$$\frac{x^3 - 3x^2 + 2x - 6}{x^2 - 2x - 3} > \frac{2}{x + 1}$$

First of all, we must write the inequality in canonical form (0 on the right-hand side). The existence conditions (denominators different from zero) will be automatically included in the sign study. We obtain

$$\frac{x^3 - 3x^2 + 2x - 6}{x^2 - 2x - 3} - \frac{2}{x + 1} > 0$$

We factor the numerator and the denominator of the first fraction, obtaining:

$$\frac{(x^2 + 2)(x - 3)}{(x - 3)(x + 1)} - \frac{2}{x + 1} > 0$$

Since we can cancel $x - 3$ and obtain an equivalent fraction, provided that $x \neq 3$ by the existence condition, we have

$$\frac{x^2 + 2}{x + 1} - \frac{2}{x + 1} > 0$$

that is,

$$\frac{x^2}{x + 1} > 0$$

The numerator is positive $\forall x \in \mathbb{R}, x \neq 0$, and the denominator is positive for $x > -1$. Combining the signs, and taking into account the condition $x \neq 3$, we obtain the solution set $(-1, 0) \cup (0, 3) \cup (3, +\infty)$.